2+1 Dimensional Gravity and Chern-Simons Theory

X. D. H.

February 10, 2026

Abstract

In this note, I'll start to learn about the Chern-Simons formulation of 2+1 dimensional gravity, following the paper of Witten (1988)[1] . For I'm reading this paper while lacking the knowledge of Basic Chern-Simons Theory and Classical 2+1 Dim Gravity, I'll include some background materials as well as preliminaries.

Contents

1 Preliminaries I: Topology and Geometry

Here I will give a brief review of some basic concepts in topology that will be useful in understanding the Chern-Simons formulation of 2+1 dimensional gravity. It mainly follows the treatment in [2] However, it is not enough or rigorous for understanding any steps in Witten's paper, but serve as an introduction to many nouns he used in the paper.

1.1 Homeomorphisms and diffeomorphisms

Consider two topological manifold X and Y. We can define a **homeomorphism** between them:

> **Definition 1.** *A **homeomorphism** is a continuous bijection $f : X \to Y$ with a continuous inverse $f^{-1} : Y \to X$.*

If such a homeomorphism exists, we say that X and Y are **homeomorphic**, denoted as $X \cong Y$. We view the two manifold as "topologically equivalent", meaning they have the same topological properties.

If the manifolds are differential manifolds, we can further define a **diffeomorphism**:

> **Definition 2.** *A **diffeomorphism** is a smooth bijection $f : X \to Y$ with a smooth inverse $f^{-1} : Y \to X$.*

If we focus one 3 dimensional manifolds, there is an importants theorem:

> **Theorem 1.** *Moise's Theorem*
> *For 2 & 3 dimentional manifolds, being homeomorphic is equivalent to being diffeomorphic.*

This theorem implies that in 2 and 3 dimensions, the topological structure and the smooth structure are essentially the same. This is not true in higher dimensions, where there exist manifolds that are homeomorphic but not diffeomorphic (exotic spheres).

1.2 Mapping Class Group

1.2.1 Isotopy

We focus on the self-diffeo of one manifold $\mathcal{M}$, $f : \mathcal{M} \to \mathcal{M}$. We can define an **Isotopy** between two diffeomorphisms:

> **Definition 3.** *An **isotopy** between two diffeomorphisms $f, g : \mathcal{M} \to \mathcal{M}$ is a continuous family of diffeomorphisms $F(s) : \mathcal{M} \to \mathcal{M}$ for $s \in [0, 1]$ such that $F(0) = f$ and $F(1) = g$. If an Isotopy exists between f and g, we say that they are **isotopic**.*

Most diffeos considered in physics are isotopic to idenity map, yet there for topologically non-trivial manifolds, there can be diffeos that are not isotopic to identity.

- eg. **Dehn Twist** on torus T^2: Cut the torus along a non-contractible loop, twist one side by 360°, and then re-glue. This operation cannot be continuously deformed to the identity map.

1.2.2 Mapping Class Group

We can define an equivalence relation using Isotopy and define its equivalent class on the set of self-diffeos of $\mathcal{M}$:

Definition 4. *The* ***Mapping Class Group*** *(MCG) of a manifold $\mathcal{M}$, denoted as MCG($\mathcal{M}$), is the group of isotopy classes of self-diffeomorphisms of $\mathcal{M}$. .*

We can prove that the equivalent class set has a group sturcture.

The mapping class group of 2 dimensional surfaces is well studies and they are generated by:

- **Dehn Twists** along non-contractible loops.

1.3 Fundamental Group

1.3.1 Definition

Definition 5. *Homotopy Curve*

Consider a series of curves satisfying: $\gamma : [0,1] \to M, \quad \gamma(0) = x_0, \quad \gamma(1) = x_0$. If two curve γ_0, γ_1 can be continuously deformed to each other while keeping the endpoints fixed, we say they are ***homotopic****, denoted as $\gamma_0 \simeq \gamma_1$.*

Mathematically, there exists a continuous map $F : [0,1] \times [0,1] \to M$ such that:

$$\begin{aligned} F(t,0) &= \gamma_1(t) \\ F(t,1) &= \gamma_2(t) \\ F(0,s) &= F(1,s) = x_0. \end{aligned} \tag{1}$$

Definition 6. *Fundamental Group*

For a fixed base point x_0 in a topological space M, the ***fundamental group*** *$\pi_1(M, x_0)$ is the set of equivalence classes of loops $[\gamma]$ based at x_0 under the homotopy relation.*

We can prove that the set of equivalence classes has a group structure, with the group operation defined as the concatenation of loops.

Theorem 2. *For connected manifolds, the fundamental group is independent of the choice of base point up to isomorphism. Therefore, we often denote it simply as $\pi_1(M)$.*

1.3.2 More on Properties

Definition 7. *Simply Connected*

A topological space M is said to be ***simply connected*** *if it is path-connected and its fundamental group is trivial, i.e., $\pi_1(M) = \{e\}$, where e is the identity element.*

Theorem 3. *Topological Invariance of Fundamental Group*

If two topological spaces M and N are homeomorphic, then their fundamental groups are isomorphic, i.e., $\pi_1(M) \cong \pi_1(N)$.

Theorem 4. *Fully Characterization of 2D Surfaces by Fundamental Group*

The fundamental group fully characterizes the topology of closed 2-dimensional surfaces. Two closed surfaces are homeomorphic if and only if their fundamental groups are isomorphic.

1.4 Covering Space

1.4.1 Covering Space Definition

We can unwrap the non-contractible loops of a manifold by considering its covering space.

Definition 8. *Covering Space*

*A **covering space** of a topological space M is a topological space $\tilde{M}$ together with a continuous surjective map $p : \tilde{M} \to M$ such that*

- *For every point $x \in M$, there exists an open neighborhood U of x such that the preimage $p^{-1}(U)$ is homeomorphic to $U \times \Lambda$, where Λ is a discrete set.*

1.4.2 Topology of Covering Space

The surjective map used by defining the covering space can induce a homomorphism between the fundamental groups of the two spaces:

Theorem 5. *Induced Homomorphism on Fundamental Groups*

Let $p : \tilde{M} \to M$ be a covering map. Then, for any base point $\tilde{x}_0 \in \tilde{M}$ with $p(\tilde{x}_0) = x_0 \in M$, the map p induces a homomorphism $p_* : \pi_1(\tilde{M}, \tilde{x}_0) \to \pi_1(M, x_0)$.

This homoporphic is injective but not necessarily surjective. The image of p_* is a subgroup of $\pi_1(M, x_0)$ called the **characteristic subgroup** of the covering space. It only consist of paths that are lifted to $\tilde{M}$ to be loops instead of open curves.

Theorem 6. *For a connected manifold, any subgroup of its fundamental group can be realized as the characteristic subgroup of some covering space.*

Thus we can classify the covering spaces with the subgroups of the fundamental group.

Definition 9. *Universal Covering Space*

*A **universal covering space** of a topological space M is a covering space $\tilde{M}$ that is simply connected, i.e., its fundamental group is trivial: $\pi_1(\tilde{M}) = \{e\}$. which corresponds to the trivial subgroup of $\pi_1(M)$.*

1.5 Quotient Space

1.5.1 Quotient Space Definition

We can construct a new manifold by quotienting a manifold with a group action.

Definition 10. *Quotient Space*

*Given a group G acting on a manifold M. The orbit of a point $x \in M$ under the action of G is the set $G \cdot x = \{g \cdot x | g \in G\}$. The **quotient space** M/G is the set of all orbits of points in M under the action of G, equipped with the quotient topology.*

We can define a natural projection from the points in M to their corresponding orbits in M/G. which gives a quotient topology by defining open sets in M/G as those whose preimages under the projection are open in M.

- In general a Quotient Space **May NOT be a Manifold**.

> **Theorem 7.** *Quotient Space as Manifold*
> *If a group G acts properly discontinuously and freely on a manifold M, then the quotient space M/G is also a manifold.*

- **freely:** For any point $x \in M$, if $g \cdot x = x$ for some $g \in G$, then g must be the identity element of G. Equivanlently the isotropy group $G_x = \{g \in G | g \cdot x = x\}$ is trivial for all $x \in M$.
- **Propery Discontinuously:**
  - Any two points in different orbits have neighborhoods that do not intersect under the group action.
  - for any point x the isotropy group $G_x = \{g \in G | g \cdot x = x\}$ is finite.
  - For any point $x \in M$, there exists an open neighborhood U of x such that for all $g \in G$ not in the isotropy group G_x, the images $g \cdot U$ do not intersect with U, i.e., $g \cdot U \cap U = \varnothing$. for all $g \in G_x$, we have $g \cdot U = U$.

> **Definition 11.** *Orbifold*
> *If a group G acts properly discontinuously but not freely on a manifold M, the quotient space M/G is called an **orbifold**.*

An orbifold is a generalization of manifold with some mild singularities.

We can find a point on M to label the orbits in M/G, the complete set of such points is called a **fundamental region** of the group action.

1.5.2 Topology of Quotient Space

> **Theorem 8.** *Topology of Quotient Space of Simply Connected Manifold*
> *If M is a simply connected manifold and G is a finite group then we can prove a relation between the fundamental group of the quotient space and the group action:*
>
> $$\pi_1(M/G) \cong G. \tag{2}$$

In this case the projection $p : M \to M/G$ is the universal covering map of the quotient space and M is the universal covering space of M/G.

> **Theorem 9.** *Connected Manifold as Quotient Space*
> *Any connected manifold M can be represented as the quotient space of its universal covering space $\tilde{M}$ by the action of its fundamental group $\pi_1(M)$:*
>
> $$M \cong \tilde{M}/\pi_1(M). \tag{3}$$
>
> *Here we define the action of $\pi_1(M)$ on $\tilde{M}$ as: Mapping the loop in M mapped to open path in $\tilde{M}$ back by identifying its endpoints.*

Mathematically, this action of $\pi_1(M)$ on $\tilde{M}$ is defined as a **Deck Transformation**:

> **Definition 12.** *Deck Transformation*
> *For any $[\gamma] \in \pi_1(M)$ we can define a map $D_{[\gamma]} : \tilde{M} \to \tilde{M}$, that satisfy $p \circ D_{[\gamma]} = p$. It gives a representation of $\pi_1(M)$ as a group of homeomorphisms of $\tilde{M}$.*

1.6 Geometization in 2D

Before we only consider topological manifolds, now we can further consider differential manifolds with riemannian metrics.

1.6.1 Standard Geometries in 2D

In 2 dimensions, there are three standard geometries with constant curvature:

- **Riemann Sphere** $\hat{\mathbb{C}} = \mathbb{C} \cup \infty$ with constant positive curvature metric:

$$ds^2 = \frac{4dzd\bar{z}}{(1+|z|^2)^2}. \tag{4}$$

- **Complex Plane** $\mathbb{C}$ with flat metric:

$$ds^2 = dzd\bar{z}. \tag{5}$$

- **Hyperbolic Plane** $\mathbb{H}^2 = \{z \in \mathbb{C} | \mathrm{Im}(z) > 0\}$ with constant negative curvature metric:

$$ds^2 = \frac{dzd\bar{z}}{(\mathrm{Im}(z))^2}. \tag{6}$$

1.6.2 Uniformization Theorem

We give out one of the most important theorems in 2D geometry:

{thm:Unif

> **Theorem 10.** *Uniformization Theorem(Pure Topology Version)*
> *Every Surface is homeomorphic (diffeomporphic) to D/Γ, where $D \in \{\hat{\mathbb{C}}, \mathbb{C}, \mathbb{H}^2\}$ and $\Gamma \subset Iso(D)$ is a properly discontinuously isometry on D.*

This is a topological statement, we only say that a topological surface can be represented as a quotient space of one of the three standard geometries. However, with suitable operations we can give the quotient space a constant curvature metric induced from the standard geometry. Thus, we know that:

- Any closed surface can be given a constant curvature metric.
- Classification of such metric = Classification of isometry group actions on the standard geometries.

2 Preliminaries II: 1st Order Formalism of Gravity

{sec:Prel

I mainly follow the discussion in [2]. For more detailed discussion and having the same notation one can refer to [3].

2.1 Vielbein and Spin Connections

We can a GR theory using a first order formalism, which use frame (aka vielbein) and spin connection as basic variables instead of metric. This is especially useful in 2+1 dimensional gravity, since it can be reformulated as a Chern-Simons theory. We define the frame fields

Definition 13. *Frame Fields are a set of orthogonal vectors in the tangent space* $e^{\mu}{}_a$ *or the inverse one form* e^a_μ*, which satisfy:*

$$g^{\mu\nu} e_\mu{}^a e_\nu{}^b = \eta^{ab}, \quad e_\mu{}^a e_\nu{}^b \eta_{ab} = g_{\mu\nu}. \tag{7}$$

Here $e^\mu{}_a$ *is the inverse of* $e_\mu{}^a$*, i.e.,*

$$e^\mu{}_a e_\mu{}^b = \delta^b_a, \quad e_\mu{}^a e^\nu{}_a = \delta^\nu_\mu. \tag{8}$$

We can show that $e^a = e_\mu{}^a dx^\mu$ forms a orthonormal basis of the cotangent space and $e_a = e^\mu{}_a \partial_\mu$ forms a orthonormal basis of the tangent space. The coeficients can also be understood as transformation matrices between coordinate basis and orthonormal basis.

This set of vectors definately forms a basis of the tangent space, thus we can consider the connections in this basis. For example, the Christoffel symbols can be written as:

$$\nabla_\mu \partial_\nu = \Gamma^\rho_{\mu\nu} \partial_\rho, \quad \text{or} \quad \nabla_\mu dx^\nu = -\Gamma^\nu_{\mu\rho} dx^\rho \tag{9}$$

Thus, a analogue is to define a connection $\omega_\mu{}^a{}_b$ as

Definition 14. *Spin Connection* $\omega_\mu{}^a{}_b$ *is defined as:*

$$\nabla_\mu e_\nu{}^a = -\omega_\mu{}^a{}_b e_\nu{}^b \tag{10}$$

Remark:

The covariant derivative here $\nabla_\mu e_\nu{}^a$ means that we only view $e_\nu{}^a$ as a dual vector in the cotangent space, with componet μ of the coordinate basis. Sometimes you may encounter the notation:

$$\nabla_\mu e_\nu{}^a = \partial_\mu e_\nu{}^a - \Gamma^\lambda_{\mu\nu} e_\lambda{}^a + \omega_\mu{}^a{}_b e_\nu{}^b \tag{11}$$

This is when we view it as a (1,1) tensor in a mixed basis, with the coordinate basis in the cotangent space and the orthonormal basis in the tangent space.

The covariant derivative of a vector is a rank (0,2) tensor, thus we know that $\omega_\mu{}^a{}_b$ transforms as a one form in the μ index. Thus:

- Both $e^a = e_\mu{}^a dx^\mu$ and $\omega^a{}_b = \omega_\mu{}^a{}_b dx^\mu$ can be understood as one forms. The inverse of the frame fields can be written as a vector field $e_a = e^\mu{}_a \partial_\mu$.
- We define the lowering and raising of the spin connection indices $\omega_{ab} = \eta_{ac} \omega^c{}_b$.

Remark:

Notice that in fact we can lower and raise the μ, ν indices with g and raise and lower the a, b indices with η. This is totally consistent with the definition of frame fields.

We can prove that if the theory is metric compatible, then the spin connection 1-form is antisymmetric in a, b indices:

$$\omega_{ab} = -\omega_{ba}. \tag{12}$$

2.2 Cartan Structure Equations

Using the above definitions, we hope to express important geometric quantities such as Riemann curvature Tensor. Moreover, we also notice that the frame fields and the spin connections are not independent, just like the Christoffel symbol can be represented by the metric. We want to:

- Represent the spin connection in terms of the frame fields or at least find a relation between them.
- Represent the Riemann curvature Tensor in terms of the spin connection.

These two goals can be achieved by the Cartan structure equations:

Theorem 11. *Cartan's first structure equation*

If the theory is torsion free, then the frame fields and the spin connection satisfy:

$$D_\omega e^a \equiv de^a + \omega^a{}_b \wedge e^b = 0, \tag{13}$$

$D_\omega e^a$ is called the gauge covariant derivative of e^a with respect to the spin connection ω.

This can be proved directly from the definition of the covariant derivative and the definition of the spin connection. The torsion free condition makes the lower indices of the Christoffel symbol symmetric, thus it cancels out when doing a exterior derivative.

Then where is the Riemann curvature tensor? It can be found in the second structure equation:

Theorem 12. *Cartan's second structure equation*

The Riemann curvature 2-form can be written as:

$$R^a{}_b = d\omega^a{}_b + \omega^a{}_c \wedge \omega^c{}_b. \tag{14}$$

Here the Riemann curvature 2-form is defined as:

$$R^a{}_b = R^a{}_{b\mu\nu} dx^\mu \wedge dx^\nu \tag{15}$$

2.3 Local Lorentz Transformations

[This is important see carrol]

Using the vielbein formalism, we can have more redundency in describing the spactime. This is because at each point we can choose different orthonormal basis in the tangent space, which are related by local Lorentz transformations.

If we use the notation of $\partial_\mu = e_\mu{}^a e_a$ and $e_a = e_a{}^\mu \partial_\mu$, then a local Lorentz transformation can be written as:

3 Preliminary III: 2+1 Dimentional Classical Gravity

{sec:}

Here I'll talk about a standard set up in studying 2+1 dimensional gravity.

3.1 Solutions to Einstein Equation in 2+1 Dimensions

3.1.1 Local Spacetime

The Riemannian curvature of 2+1 dimensional gravity can be written directly in terms of Ricci Tensor and the metric:

$$R_{\mu\nu\rho\sigma} = g_{\mu\rho}R_{\nu\sigma} + g_{\nu\sigma}R_{\mu\rho} - g_{\nu\rho}R_{\mu\sigma} - g_{\mu\sigma}R_{\nu\rho} - \frac{1}{2}(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho})R \tag{16}$$

Thus if we consider the vaccum Einstein equation $R_{\mu\nu} = 0$, the Riemannian curvature vanishes identically! And if we consider the cosmological constant case $R_{\mu\nu} = \Lambda g_{\mu\nu}$, the Riemannian curvature becomes:

$$R_{\mu\nu\rho\sigma} = \Lambda(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho}) \tag{17}$$

This is a Riemann Tensor of a constant convature spacetime.

- When $\Lambda = 0$, the spacetime is locally flat, i.e., locally isometric to Minkowski space $\mathbb{R}^{2,1}$.
- When $\Lambda > 0$, the spacetime is locally de Sitter space dS_3.
- When $\Lambda < 0$, the spacetime is locally anti-de Sitter space AdS_3.

3.1.2 Global Spacetime

However, being locally constant curvature spacetime doesn't mean that the theory is trivial, it may include various global properties. However, we shall ask what kinds of global topoloigy may admit a constant curvature metric as well as a lorentzian structure(for physical spacetime is lorentzian).

The answer is given by the following theorem:

> **Theorem 13.** *Let M be a compact 3 manifold with a flat, time-orientable Lorentzian metric and a purely spacelike boundary. Then M has the topology of:*
>
> $$M \cong [0,1] \times \Sigma \tag{18}$$
>
> *Here Σ is a closed surface homeomorphic to one of the boundary components of M.*

Thus the topology of a physical spacetime is fixed by the topology of a initial spacelike slice Σ. In the following context we shall only focus on this kind of spacetime topology.

3.2 Naive Phase Space Analysis (Witten)

3.2.1 Discussion Set Up

How to quantize a Theory, we usually start from analyzing the phase space of the theory and then construct a poisson braket of the phase space. Then one can transform the poisson bracket into commutator and get the quantum theory. Thus to study the quantum gravity in 2+1 dimensions, we need to first learn about the phase space of the classical 2+1 dimensional gravity.

The phase space of a classical theory is often defined as:

Definition 15. *Phase Space*
The phase space is the space of all $q, \dot{q}$ at $t = 0$ surface.

However, this definition is not covariant and we need tons of constraints to make it work, thus in 2+1 dimensional gravity, we use the following definition:

Definition 16. *Phase Space*
The phase space is the space of all classical solutions to the equations of motion Modulo gauge transformations.

For 2+1 dimensional gravity, this means that we want to find all the solutions to:

$$R_{\mu\nu} = 0 \tag{19}$$

To do this, we might first assume a spacetime topology of the manifold we want to consider, we focus on a manifold:

- With topology $M \cong [0, 1] \times \Sigma$. where Σ is a genus g closed surface.

We now simplify the problem of analyzing the phase space of 2+1 dimensional gravity to finding the space of all solutions to the vacuum Einstein equation $R_{\mu\nu} = 0$ on a manifold with topology $M \cong [0, 1] \times \Sigma$. These solutions can be found by quotienting the Minkowski space by a discrete subgroup of the isometry group of Minkowski space, which is the Poincare group $ISO(2, 1)$. In Witten's paper he gives out a analysis of those manifolds.

3.2.2 Manifolds with initial & final singularities

Witten explicitly constructs these class of solutions as examples, by quotienting the light cone of Minkowski, now we explain the procedure:

- **Preparation: Quotienting $\mathbb{H}$ to Σ_g**

  We can construct a 2-manifold with genus $g \geq 2$ by quotienting the hyperbolic plane.

  - **Step 1:** We first find the fundamental group of the surface Σ_g which is $\pi_1(\Sigma_g)$.
  - **Step 2:** We find a subgroup of $PSL(2, \mathbb{R})$ naming Γ which is isomorphic to $\pi_1(\Sigma_g)$ and acts **discretely** on $\mathbb{H}$.
  - **Step 3:** The quotient $\mathbb{H}/\Gamma$ defines a Riemann surface with genus g and a metric with constant negative curvature.

[In fact I don't understand how this step works mathematically, eg how to construct the metric and why we don't require the group to be free but only properly continuous (discrete), etc. I need to learn more about this step.]

Remark:
Note that here we quotient by a discrete subgroup other than a properly discontinuous subgroup, because the action of a properly discontinuous subgroup is equivalent to being discrete in the case of being a subgroup of $Aut(\mathbb{H}) = PSL(2, \mathbb{R})$.

Then we apply a similar procedure to the light cone of Minkowski space to construct a flat 3-manifold with topology $M \cong [0, 1] \times \Sigma_g$ with initial singularity and final singularity.

To do this we first list out some useful facts about the 2+1 Minkowski spacetime:

- Normally we have the metric of:

$$ds^2 = -dt^2 + dx^2 + dy^2 \tag{20}$$

And the light cone is defined by:

  - X^+ future light cone: $t^2 > x^2 + y^2, t > 0$
  - X^- past light cone: $t^2 > x^2 + y^2, t < 0$

- The isometry group is the 3 dimensional Poincare group $ISO(2,1)$ which is the semidirect product of the Lorentz group $SO(2,1)$ and the translation group $\mathbb{R}^{2,1}$.

  - The proper orthochronous Lorentz group $SO^+(2,1)$ is isomorphic to $PSL(2,\mathbb{R})$.

Quotienting the light cone to get a 3-manifold with initial & final singularities

We first notice that the light cone can be made up by hyperbolic planes:

$$t^2 = x^2 + y^2 + \tau^2 \tag{21}$$

where τ is a parameter for the hyperbolic plane.

- Each hyperbolic plane is isometric to $\mathbb{H}$ and the isometry group of the hyperbolic plane is the Lorentz group $SO^+(2,1)$ which is a subgroup of the Poincare group $ISO(2,1)$

Thus given a Σ_g we can construct a subgroup of the lorentz group Γ isometric to $\pi_1(\Sigma_g)$ and quotient the hyperbolic plane by this subgroup to get a Σ_g. If we do this quotient for the whole X^+ light cone we can get a 3 manifold with topology $M \cong [0,1] \times \Sigma_g$ and with an initial singularity at $\tau = 0$.

- The resulting manifold is a solution to the vaccum Einstein Field Equation for it is contructed by quotienting an isometry of Minkowski spacetime. This quotient group preserves the metric thus the resulting manifold is still flat. [I don't understand why??]

- The resulting manifold has a $\Sigma_g \times [0,1]$ topology. This is because the lorentz group act the hyperbolic plane to itself due to its feature of preserving Minkowski metric. Thus, Γ act independently on each hyperbolic plane. Moreover, the isometry group of the hyperbolic plane is the lorentz group and the hyperbolic plane is isometric to $\mathbb{H}$. Thus, quotienting the hyperbolic plane with the subgroup Γ of the lorentz group is equivalent to quotienting $\mathbb{H}$ with the same subgroup, which gives us a Σ_g. Thus, the resulting manifold has a $\Sigma_g \times [0,1]$ topology.

Finally, we can count how many distinct manifold can be constructed by this procedure, the answer depends on the choise of Γ subgroup. The final answer is that the moduli of genus g surface, which is $\mathbb{R}^{6g-6}$ gives us the label of distinct manifolds we can construct by this procedure. [I don't understand this counting, it need some knowledge of Teichmuller space and moduli space of Riemann surface.]

3.2.3 General Flat Manifolds

Now we try to construct more general flat manifolds with topology $M \cong [0,1] \times \Sigma_g$ by quotienting more general subspace of the Minkowski space.

4 Chern-Simons Theory and Gravity (Witten)

{sec:Cher

Here we continue on following Witten's discussion on the relation between 2+1 dimensional gravity and Chern-Simons theory [1].

4.1 Review on Chern-Simons Theory

4.2 1st Order Formalism of 3D Gravity

4.3 On-Shell Diffeomorphism as Gauge Transformation

5 Chern-Simons Theory and AdS3 (AGR)

{sec:Cher

Modernly, we often use another way to get a Chern-Simons Theory from the 3D Gravity with non zero cosmology constant Λ or if we focus on AdS3 then the AdS length l. This approach show that pure gravity in AdS3 can be equivalent to two Chern-Simons action. This section we maily follow the lecture notes [4].

5.1 Double Chern-Simons Theory as AdS3

5.2 On-Shell Diffeomorphism as Gauge Transformation

6 Quantization and Phase Space Analysis (Witten)

{sec:Quan

References

[1]E. Witten, “2 + 1 dimensional gravity as an exactly soluble system”, Nuclear Physics B **311**, 46–78 (1988).

[2]S. Carlip, *Quantum gravity in 2+1 dimensions*, Cambridge Monographs on Mathematical Physics (Cambridge University Press, Cambridge, 1998).

[3]S. M. Carroll, *Spacetime and geometry: an introduction to general relativity*, Pearson new international edition (Pearson Education, Harlow, 2014), 513 pp.

[4]G. Compère and A. Fiorucci, *Advanced Lectures on General Relativity*, (Feb. 8, 2019) arXiv:`1801.07064 [gr-qc, physics:hep-th]`, `http://arxiv.org/abs/1801.07064` (visited on 03/13/2024), pre-published.

Todo Notes / Open Points